LAB 2

VLAN Configuration & Network Segmentation

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LAB OVERVIEW

TASKS

1. Configure VLAN 10 (Students) and VLAN 20 (Faculty) on switches
2. Assign switch ports to the respective VLANs
3. Set up subinterfaces on the router to enable inter-VLAN routing
4. Verify VLAN communication by testing connectivity within VLANs and between VLANs

HARDWARE

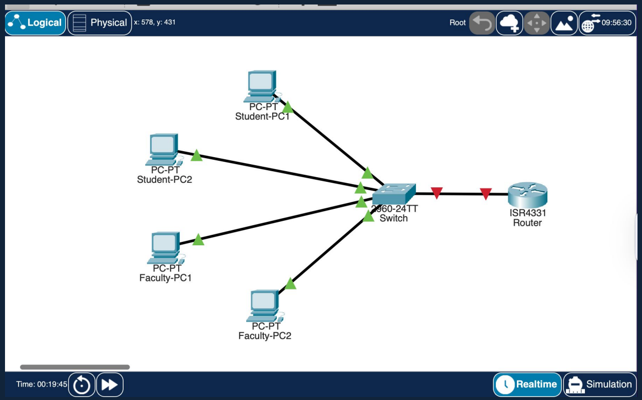
- 1X Cisco ISR4331 Router
- 1X Cisco 2960-24TT Switch
- 4X PC End Devices
- 5X Straight-Through Cables

SOFTWARE

CISCO Packet Tracer

VLAN ASSIGNMENT

Text



VLAN Port & IP Assignment Table

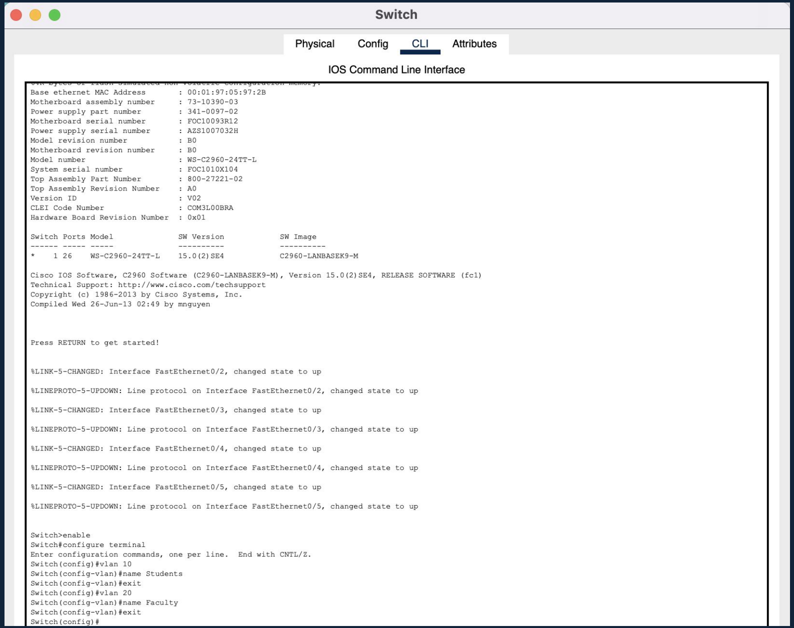
DEVICE	VLAN	IP ADDRESS	PORT
Student-PC1	VLAN 10	192.168.10.10	Fa0/2
Student-PC2	VLAN 10	192.168.10.11	Fa0/3
Faculty-PC1	VLAN 20	192.168.20.10	Fa0/4
Faculty-PC2	VLAN 20	192.168.20.11	Fa0/5

Subnet Mask: 255.255.255.0 · VLAN 10 GW: 192.168.10.1 · VLAN 20 GW: 192.168.20.1

Each device was assigned a unique static IP within its VLAN subnet, with the default gateway pointing to the corresponding router subinterface: 192.168.10.1 for Students and 192.168.20.1 for Faculty

SWITCH VLAN CONFIGURATION

Switch — CLI Interface



Create VLANs 10 & 20

Entered global config mode using configure terminal command and created VLAN 10 named "Students" and VLAN 20 named "Faculty" using the vlan and name commands.

PORT-TO-VLAN ASSIGNMENTS

Ports Assigned to VLAN 10 (Students)

```
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
```

Ports Assigned to VLAN 20 (Faculty)

```
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#internet fastEthernet 0/5

Switch(config)#interface fastEthernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
```

For VLAN 10 (Students): interfaces Fa0/2 and Fa0/3 were configured with switchport mode access followed by switchport access vlan 10, placing Student-PC1 and Student-PC2 exclusively in the Students segment.

For VLAN 20 (Faculty: interfaces Fa0/4 and Fa0/5 were configured with switchport mode access followed by switchport access vlan 20, placing Faculty-PC1 and Faculty-PC2 exclusively in the Faculty segment.

TRUNK CONFIG & VLAN VERIFICATION

Trunk Port (Fa0/1 → Router)

```
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

show vlan brief Output

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Students                         active    Fa0/2, Fa0/3
20   Faculty                          active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Trunk Port - Router

FastEthernet 0/1 was set to switchport mode trunk telling the switch to tag outgoing frames with 802.1Q VLAN IDs so it can identify which subinterface to forward each frame through.

VLAN 10 — Students Confirmed

The show vlan brief output confirms VLAN 10 "Students" is active with Fa0/2 and Fa0/3 assigned. This verifies Student-PC1 and Student-PC2 are correctly isolated within the Students segment at the switch level.

VLAN 20 — Faculty Confirmed

VLAN 20 "Faculty" is confirmed active with Fa0/4 and Fa0/5 assigned. Faculty-PC1 and Faculty-PC2 are correctly placed in the Faculty segment.

ROUTER SUBINTERFACE CONFIG.

Subinterface for VLAN 10

Created subinterface G0/0/0.10, set encapsulation dot1Q 10 so the router recognizes VLAN 10-tagged frames, then assigned IP 192.168.10.1/255.255.255.0.

Subinterface for VLAN 20

Created subinterface G0/0/0.20 with encapsulation dot1Q 20 and IP 192.168.20.1/255.255.255.0. This gateway IP enables Faculty PCs to reach the router and communicate with devices in the Students VLAN via inter-VLAN routing.

show ip route Command

Output showed both VLAN networks as directly connected through the router subinterfaces.

Router-on-a-stick Set Up

VLAN 10 Subinterface

```
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
```

VLAN 20 Subinterface

```
Router(config)#interface gigabitEthernet 0/0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
```

show ip route output

```
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0.20
```

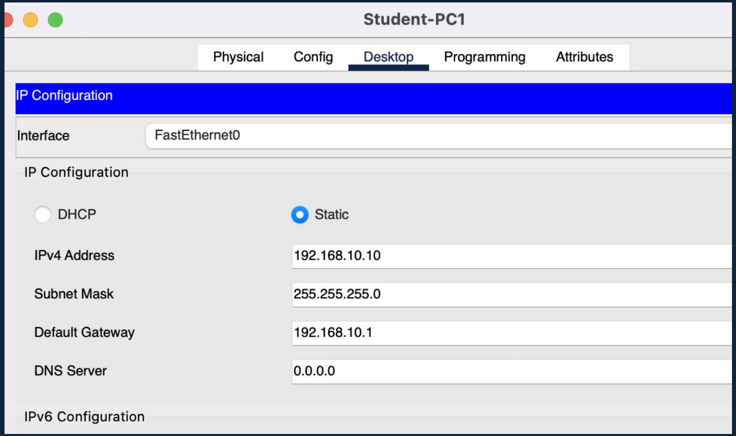
PC STATIC IP CONFIGURATION

Each PC was manually configured with a static IPv4 address within its respective VLAN subnet. Student PCs use the 192.168.10.0 network and Faculty PCs use 192.168.20.0. The default gateway was set to the corresponding router subinterface IP to ensure proper inter-VLAN communication.

DEVICE	IP ADDRESS	GATEWAY	VLAN
Student-PC1	192.168.10.10	192.168.10.1	**VLAN 10**
Student-PC2	192.168.10.11	192.168.10.1	**VLAN 10**
Faculty-PC1	192.168.20.10	192.168.20.1	**VLAN 20**
Faculty-PC2	192.168.20.11	192.168.20.1	**VLAN 20**

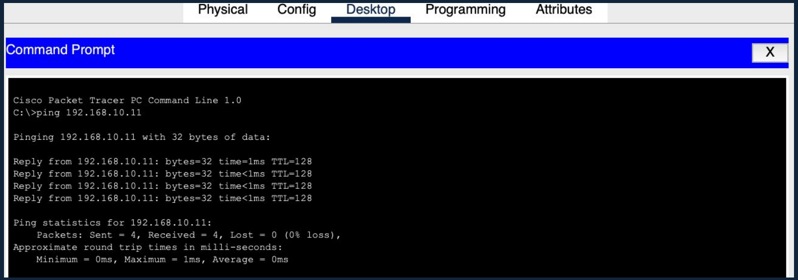
Subnet Mask: 255.255.255.0 · 192.168.10.0/24 (Students) · 192.168.20.0/24 (Faculty)

Student-PC1 — IP Configuration Panel

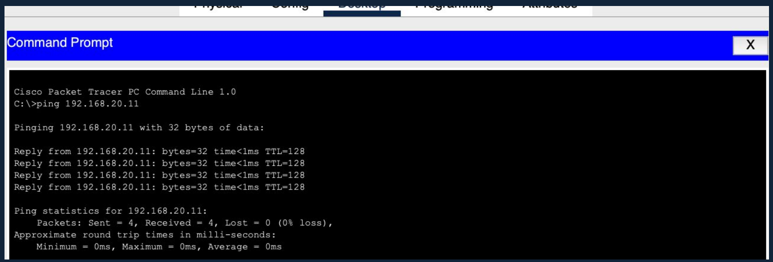


CONNECTIVITY TESTING

Ping Within VLAN 10 (Students)



Ping Within VLAN 20 (Faculty)

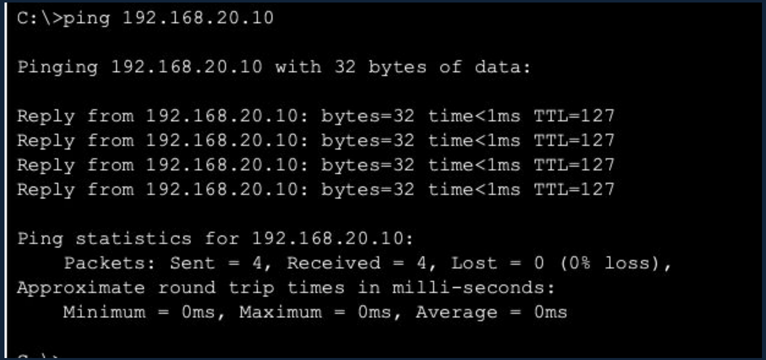


4/4
Packets Received

0%
Packet Loss

3/3
Hosts Reached

Ping Between VLANs (Inter-VLAN Routing)



CHALLENGES FACED & SOLUTIONS

⚠️ **CHALLENGE**

Forgot to set port Fa0/1 to trunk mode, so only one VLAN's traffic was passing through.

✓ **SOLUTION**

Used show interfaces trunk to verify, then ran switchport mode trunk to fix it.

⚠️ **CHALLENGE**

Forgot to add the encapsulation dot1Q command on the router subinterfaces, so the router wasn't recognizing VLAN traffic.

✓ **SOLUTION**

Matched each subinterface's dot1Q encapsulation to the correct VLAN ID on the switch.

⚠️ **CHALLENGE**

The first ping between VLANs showed packet loss and timed out.

✓ **SOLUTION**

Learned this is normal – It was just ARP resolution. The following pings all came back with 0% packet loss.

LESSONS LEARNED

01

VLANs Separate Networks

VLANs typically allow a single physical switch to be spread into multiple logical networks. This helps keep different groups like students and faculty in separation while still using the same hardware.

02

Trunk Ports Carry Multiple VLANs

A trunk port allows traffic from multiple VLANs to travel over one connection between devices like a switch and a router. This is important for sending tagged VLAN traffic to the router for routing.

03

Router-on-a-Stick Enables Inter-VLAN Communication

Router subinterfaces with dot1Q encapsulation allow devices in different VLANs to communicate. The separate VLAN uses its own subinterface and gateway IP address so the router can route traffic between them.

REAL-WORLD APPLICATIONS

Separating Different Users Within an Organization

Organizations such as universities, offices, and hospitals may need to separate different groups of users while using the same physical network.

Hospitals

Isolate sensitive patient information, medical records, and preventing unauthorized access to comply with HIPAA

Universities

Separating network traffic between students, faculty, and administrative staff to improve network security, avoid data breaches, and protect sensitive personal information